Modelling 50-Year Survival Among University of Adelaide Graduates

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1 Summary

This report explains the effects of faculty, sex and year of graduation in determining the probability of surviving for 50 years after graduation, using binomial models. Section 3 reveals notable patterns, particularly the effect of sex on survival probability, with females generally much more likely to survive for 50 years after graduation. It also demonstrates how those from the science and engineering faculties had higher observed 50-year survival proportions than those from the medicine or arts faculties. When considering models with all datapoints, our models confirm the variables sex and faculty are the most influential predictors, whilst year of graduation shows negligible significance.

In section 5.1 we investigate reducing the parameter number, more specifically removing the year variable and combining factors. We demonstrate that combining the faculties of science and engineering as well as medicine and arts demonstrate no significant drop in explanatory power, whilst reducing parameter number. Investigating interaction effects as well as binning our `year` variable into discrete categories has no statistically significant additional effect on explanatory power when compared to not including those parameters in the first place. When modelling our entire dataset, we find that `model4` found in fig. 10 was preferred for its simplicity and adequate fit.

When separating the data by sex, we use 40 grouped observations (38 with nonzero totals) to select a male specific model, found in fig. 18 which strikes a balance between the number of predictors and explanatory power. When considering female only data, we do not find sufficient evidence that any of the predictors, be that `faculty` or `year` have enough explanatory power to be preferred to the null model at a 5% significance level.

2 Introduction

This report investigates the effect of various factors, for example sex, university faculty and year of graduation on a person's chances of survival for 50 years after graduation. Data has been collected by the University of Adelaide, Australia from university students graduating between 1938 and 1947. Our goal is to identify which variables have an effect on survival, by systematically evaluating and comparing various binomial models. Through model selection and comparison we aim to get an efficient and interpretable way of understanding survival among university graduates.

3 Exploratory Data Analysis

The data record the number surviving, `survive`, out of the group total, `total`, making a binomial model appropriate. The data consists of 60 observations across five variables: `year`, `faculty`, `sex`, `survive`, and `total`. The year of graduation spans from 1938 to 1947, with 6 observations for each year. The `faculty` variable appears to be a categorical variable, containing 4 levels: A, E, M, S, corresponding to the faculties arts, engineering, medicine and sciences respectively. Arts has 20 observations, engineering has 10, medicine has 10, and sciences has 20 observations.

The variable `sex` is also a categorical variable with 2 levels: M and F, with 40 and 20 observations respectively. To explore these differences, we plot the proportion surviving in each group, calculated as `survive/total`.

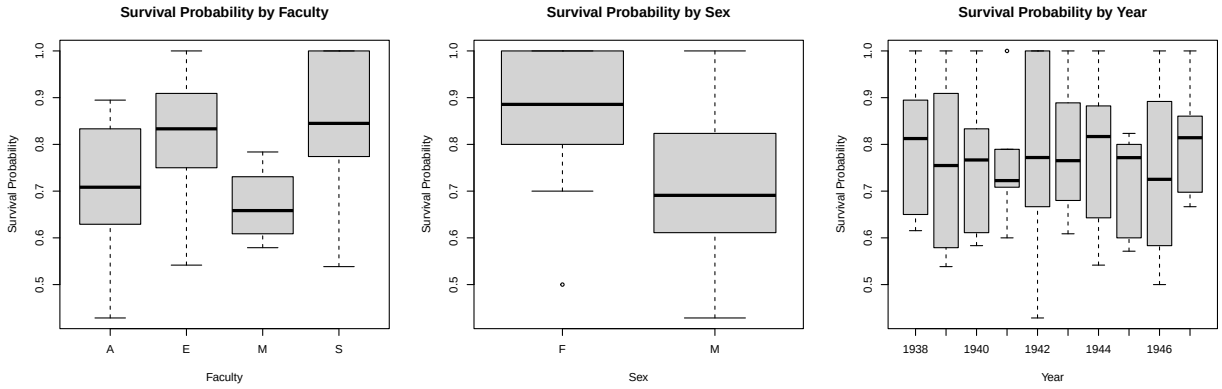


Figure 1: Probability of Survival by Faculty, Sex and Year

Figure 1 shows us the distributions of mortality observations across different faculties, sexes and years. Sciences has the highest median survival probability, with each quartile being higher than the other faculties, indicating consistently high survival probabilities. Medicine on the other hand displays the lowest median survival probability, with a small interquartile range indicating observations are more tightly clustered around the median. Engineering shows a high median with again a narrow interquartile range, however a large range. Arts displays more variation and a lower median than Sciences and Engineering. We can also see from visual inspection that sex appears to be associated with survival, with females having much higher survival rates when compared to males, with a smaller range. The survival probability by year fluctuates with no clear visual trend. There are at most 6 nonempty groups per box however, meaning the results may be sensitive to individual variation and not fully representative, we can see no definite trend.

4 Mathematical Basis for Investigation

Within this investigation, we will be using a number of measures to assess both the fit of a model as well as the significance of parameters. Firstly to assess model fit, we will compute the deviance (D). Let our model have a vector of fitted parameters $\hat{\boldsymbol{\theta}}$ with corresponding fitted values $\hat{\boldsymbol{\mu}}$. We also have the saturated model with parameters $\tilde{\boldsymbol{\theta}}$ and fitted values $\tilde{\boldsymbol{\mu}}$. Further let the observed data be $\boldsymbol{y} = [y_1, \ldots, y_n]$ where each y_i denotes the number of successes out of m_i trials. The deviance of the model is twice the difference between the log-likelihood of the saturated model and the log-likelihood of our model. That is:

$$D = 2 \cdot \left\{ l(\tilde{\boldsymbol{\theta}}; \boldsymbol{y}) - l(\hat{\boldsymbol{\theta}}; \boldsymbol{y}) \right\} = 2 \sum_{i=1}^{n} \left\{ y_i \log\left(\frac{y_i}{\hat{\mu}_i}\right) + (m_i - y_i) \log\left(\frac{m_i - y_i}{m_i - \hat{\mu}_i}\right) \right\}.$$

To compare two nested models (i.e. a model M_0 contained within a more complex model M_1), we perform a likelihood ratio test using the `ANOVA` function. This tells us if there is a significant difference in explanatory power between both models, so whether the extra parameters in the more complex model are significant over the simpler model. Consider two nested models M_0 and M_1 with r_0 and r_1 parameters respectively such that $r_0 < r_1$. Let them both have deviances D_0 and D_1 respectively. The `ANOVA` function with the `Chisq` test automatically performs a likelihood ratio test, that is calculate the test statistic $D_0 - D_1$, which is approximately $\chi^2_{r_1 - r_0}$ under the null hypothesis, finally outputting a p-value.

Alongside these tests for model fit and model comparison, we also discuss the significance of individual parameters. Let us define our vector of fitted parameters as $\hat{\boldsymbol{\beta}}$, with standard errors denoted $SE(\hat{\boldsymbol{\beta}})$. If a model has r parameters, define:

$$z_i = \frac{\hat{\beta}_i}{SE(\hat{\beta}_i)}, \quad \forall i \in \{1, \ldots, r\}.$$

Under the null hypothesis $\beta_i = 0$, z_i is approximately $N(0, 1)$. Thus for each parameter, we can calculate $p_i = 2 \cdot \Phi(-|z_i|)$ where Φ is the CDF of the standard normal distribution. Where p_i values are small, we have evidence that $\beta_i \neq 0$. If p values are large, we do not reject the null hypothesis that $\beta_i = 0$.

Within this project, we have also chosen to use the logit link function, this relates the survival probability p_i to the linear predictor η_i through the relation:

$$\eta_i = \text{logit}(p_i) = \log\left(\frac{p_i}{1 - p_i}\right).$$

5 Model Fitting

We begin by considering all three predictors together, so that each association with survival is assessed while accounting for the other two. From section 2 we have determined that we have a binomial model. Beginning with a basic model of `year + faculty + sex`, fig. 6 shows that the reference levels are set as the faculty of arts and the sex as female. Inspection of the summary shows that the probability of survival are significantly different between the arts and engineering faculties and the arts and science faculties, namely science and engineering graduates have a higher probability of survival when compared to graduates from the arts faculty, both with p-values of less than 1%. Furthermore we can see that sex also shows significance with the `sexM` parameter having a p-value of 0.000136 and an estimate of -0.85542, indicating that men have a lower chance of survival for 50 years after graduation when compared to women. Whilst the intercept has a large negative value, it also has a large standard error, and thus a low significance, furthermore the year of graduation does not show a statistically significant association with the chances of survival for 50 years after graduation. The parameter `facultyM` having an estimate of 0.18391 indicates that people from the faculty of medicine have a slightly higher chance of survival for 50 years after graduation when compared to the faculty of arts, however with a p-value of 0.372042, there is not enough evidence for this to be true at a 5% significance level.

The analysis of deviance table, also located in fig. 6, supports these observations. When adding `year` to the null model, deviance only drops slightly by 2.3427, suggesting that year of graduation does not meaningfully improve model fit. The addition of `faculty` reduces the deviance by 9.364, indicating a more substantial effect on survival probability prediction. The addition of `sex` as a parameter drops the deviance by a significant amount (15.4856), and with 1 degree of freedom difference, provides evidence that it is an extremely powerful predictor.

To illustrate how the fitted model is used, we estimate survival probabilities for a male medicine graduate from 1941 and a female engineering graduate from 1938, including a combination not observed in the data. With this model, fig. 7 shows the predictions of our models with the example data:

```
[1] sex = M, year = 1941, faculty = M
[2] sex = F, year = 1938, faculty = E
```

We get predictions: 0.6694586 and 0.8749723 respectively, whilst there is data for the mortality of males in the faculty of medicine who graduated in 1941, there is not data for females graduating in

1938 in the faculty of engineering. This means that the second prediction relies on extrapolation from the fitted model rather than direct observation, which may carry greater uncertainty and carries a larger standard error. In contrast the first prediction is interpolated from the model, so will likely carry less uncertainty as a prediction and has a lower standard error.

5.1 Model Simplification

Having fitted the initial model, we now consider whether all of its terms are needed to describe the observed survival patterns. Since the `year` parameter in our previous model demonstrated itself to be insignificant, we can attempt to simplify the model using the code in fig. 8, this removes `year` as a parameter. The results of this simplified model show that the removal of `year` does not significantly worsen the model fit. A likelihood ratio test comparing `model1` (with year) and `model2` (without year) returns a p-value of 0.2668, as shown in the ANOVA table. This indicates that the difference in deviance between the two models (1.2332) is not statistically significant, and therefore, the more complex model does not provide a significantly better fit to the data. Furthermore, whilst `model2` does have a higher deviance to `model1`, it has a lower AIC, suggesting that the simpler model is preferable when balancing model fit with complexity. Figure 2 shows the diagnostic plots for both `model1` and `model2`. The residuals vs fitted plots for both models indicate a reasonably good spread of residuals, with no strong patterns. `model2` shows slightly more clustering of residuals around certain fitted values, however this is likely a consequence of having the model based on entirely categorical variables with few factor levels. The Q-Q plots do not appear dramatically different between models either. Although `model1` shows marginally better residual behaviour, the difference is not substantial. Given the statistical insignificance of the `year` parameter and the slightly better AIC for `model2`, we conclude that the simplified model does not result in a significant loss in explanatory power when compared to `model1`.

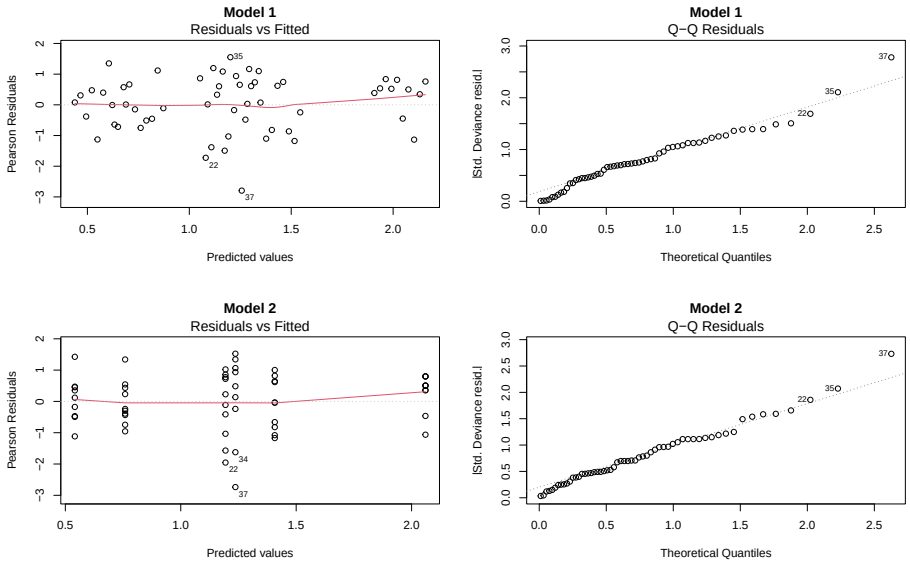


Figure 2: Model 1 vs Model 2 fit

Now we have `model2` as our efficient baseline, looking at the summary in fig. 8, we can see that there is little evidence of a difference between the medicine and arts faculties. With this information, fig. 9 shows how we can combine these factor levels to reduce the parameter number by merging the medicine (`M`) and arts (`A`) faculties into a new level, `MA`, while retaining engineering (`E`) and science (`S`) as separate levels. The results show that our baseline (reference) faculty is now engineering, as it does not appear on the summary, the model contains a slightly higher residual deviance, which is to be expected as the parameter number has decreased. More notably however the AIC has continued to decrease, with a new AIC of 204.01 in `model3` as compared to 204.87 in `model2`, showing that its

efficiency has increased with respect to parameter number. If we once again perform a likelihood ratio test between `model2` and `model3` (results in fig. 9), we can see that there is no significant loss of explanatory power when combining the `M` and `A` factor levels, with a p-value of 0.2846. This provides no significant evidence that our new model simplification reduces explanatory power. Figure 3 shows us a visualisation of the model fit between models 2 and 3. The diagnostic plots comparing models 2 and 3 demonstrate that the simplified `model3` performs comparably to `model2` in terms of residual behaviour, despite having fewer parameters. In the Residuals vs Fitted plots, both models show residuals scattered fairly symmetrically around zero with no major residual issues. The Q-Q Residuals plot shows `model3` with datapoints closer to the theoretical quantile line in the upper quantiles, and in the middle and lower quantiles performing around the same. This supports the conclusion that the simplification introduced in `model3` maintains an adequate model fit while improving interpretability and efficiency.

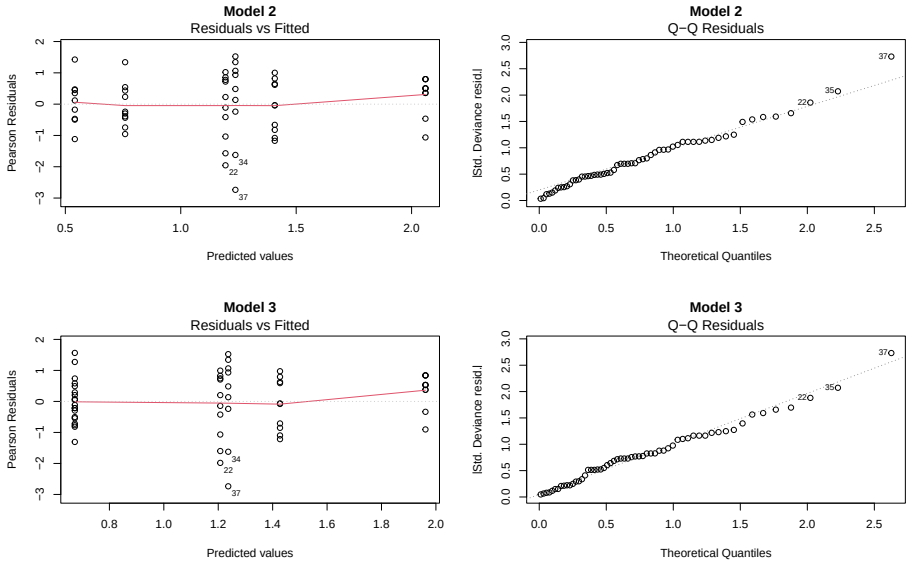


Figure 3: Model 2 vs Model 3 fit

As well as confirming a good model fit for the new `MA` factor, the summary in fig. 9 also shows us that there doesn't appear to be a significant difference between the engineering and science faculties at predicting mortality. As with model 3, we can combine these two factors. Figure 10 shows how we have combined these factors, the new summary table states that each parameter is extremely significant, each with p-values less than 1%. These results indicate that individuals in the science or engineering faculties have a significantly higher probability of survival compared to those in either the medicine or arts faculties, and males have a significantly lower probability of survival than females. As well as the significance of each parameter, we once again have a decreasing AIC value of 202.03 in `model4` as compared to 204.01 in `model3` showing improved parameter efficiency of the model. Additionally performing a likelihood ratio test indicates that `model4` is not significantly different from `model3`. Before we start adding parameters, or investigating interaction terms, fig. 11 shows us a likelihood ratio test between `model4` and our original `model1`, which provides evidence for no significant difference between explanatory power between the two models despite having a much lower parameter number.

5.2 Investigating Interactions

We next check whether allowing the effects of faculty group, sex or graduation year to vary with one another improves the model. So far, we have focused on removing variables to simplify the models, without considering whether the effect of one variable depends on another. For example, the effect of sex on survival may differ across faculties. Continuing from `model4`, we can add an

interaction term between sex and our combined faculties `MA` and `SE`. Figure 12 shows the summary table as well as the ANOVA comparison between `model4` and the new `model5` including interactions between faculty and sex. Ignoring for now the significance of each parameter individually, we can see that the AIC increases as we include the interaction term, furthermore ANOVA analysis shows no significant difference in explanatory power between each model. Thus we can conclude the introduction of this new variable is not necessary. For the interactions involving graduation year, we compare each model with a model retaining faculty group, sex and year as main effects:

```
year_baseline = glm(cbind(survive, total-survive) ~ faculty3 + sex + year,
                    family = binomial)
sex_year_model = glm(cbind(survive, total-survive) ~ faculty3 + sex*year,
                     family = binomial)
faculty_year_model = glm(cbind(survive, total-survive) ~ faculty3*year + sex,
                         family = binomial)
anova(year_baseline, sex_year_model, test = 'Chisq')
anova(year_baseline, faculty_year_model, test = 'Chisq')
```

The likelihood ratio tests give p-values of 0.141 for the sex:year interaction and 0.602 for the faculty-group:year interaction, providing no significant evidence of interactions with graduation year. In conclusion, these tests provide no significant evidence for the interactions considered.

5.3 Binned Time Effects

A weak linear association with graduation year could still conceal differences between periods, so we also examine grouped years. Whilst the `year` parameter is not included in our most recent model (`model4`), it is currently treated as a continuous parameter as supposed to a discrete categorical feature. To explore whether converting this feature into discrete bins adds explanatory power, we will begin by testing various models. One notable occurrence in the time-frame of our data is World War II. We can categorise our `year` variable into 3 bins: pre-war (1938-1939), during-war (1940-1945) and post-war (1946-1947). Building from our current optimal model, `model4`, we can now add this new factor variable on top to see if it adds any more explanatory power. Figure 13 shows us the summary and ANOVA table. Whilst the model does have the lowest AIC thus far at 201.59, a likelihood ratio test when compared to `model4` returns a p-value of 0.1088 indicating there is no significant difference in explanatory power when compared to `model4`. Figure 4 shows us a boxplot of survival probabilities based on these bins. By visual inspection we cannot see explicit drastic changes in survival probabilities between bins, we have a slightly higher range in the 'During-War' bin, however this may be caused by the fact that that category has more data than the other two.

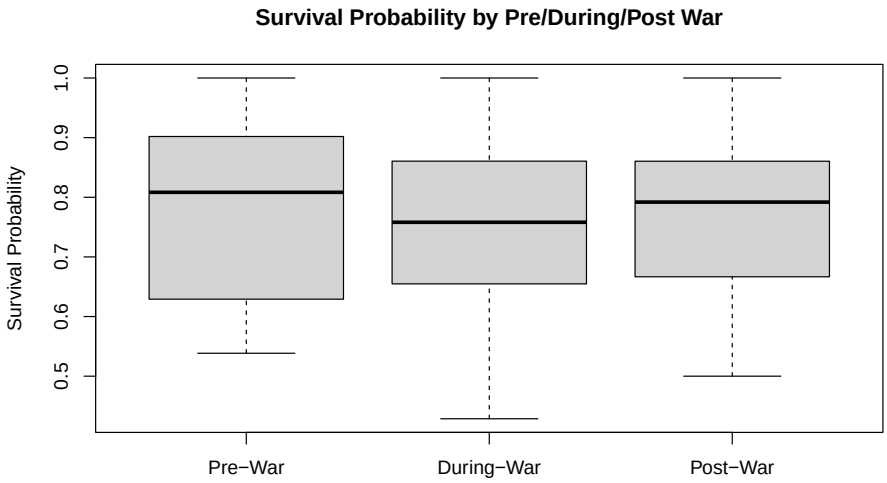


Figure 4: Probability of survival by year, organised by graduation pre, post and during WWII.

Another choice of binning is before 1940 and from 1940 onwards, creating 2 bins. Figure 14 once again shows the summary and ANOVA table of this model. We can see that based on the likelihood ratio test with `model4`, we have a model will very little significance in terms of difference from `model4`. From the summary table also we have a larger AIC than `model4` and our previous `war_model`, indicating that it underperforms in just about every respect. It has a marginally better residual deviance as compared to `model4`, however this is to be expected as it has a higher parameter number.

Overall we can determine that choosing bins to discretise our `year` variable provides no significant boost to explanatory power when compared to the simpler `model4`. If you want the lowest AIC among the models considered, you may want to use the `war_model`, however the likelihood ratio test in fig. 13 tells us this is not a significant increase in explanatory power.

6 Separating Male and Female Data

So far we have not treated male and female data separately, even though in section 5.2 we did see if there was an interaction term between faculty and sex, this was completed after we had combined factors in section 5.1. This combining of factors may not be relevant to both sexes, especially if we have an unequal number of entries for men and women as we do.

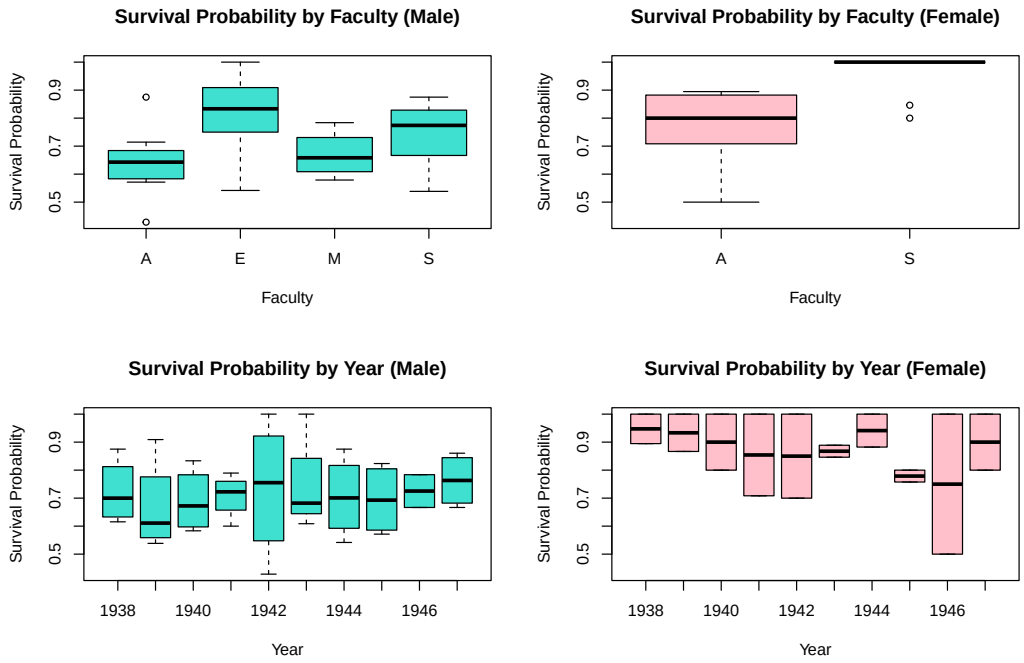


Figure 5: Probability of survival by faculty and year, grouped by sex.

Figure 5 shows the survival probabilities by faculty and year, grouped by sex. The boxplot shows notable differences between male and female students, not only because of the absence of female data for the engineering and medicine faculties, but in the contrasts of where data lies for both genders. For example, across the arts and sciences faculties, females appear to show higher survival probabilities, this is also true for the data across all graduation years, females appear to have higher survival probabilities. Previously, in fig. 1, we could see visual evidence for the combining of factors, however when looking at the female plots within fig. 5, we cannot see much evidence for why you would want to combine these faculties. Figure 5 does give some evidence for the merging of factors within the male faculties, however we cannot determine if this is significant just from visual inspection.

6.1 Male Data

We will first start by analysing only male data. Similar to `model1`, we will consider the model with `faculty + year` at first. Figure 15 shows the summary table and ANOVA table for our first male only model. From the summary table, we can see that only two parameters have a p-value of less than 5%, so the remaining three coefficients are not individually significant at this level. This does not establish that their effects are absent. Analysis of the ANOVA table suggests that once again the `year` parameter is providing the least amount of explanatory power. Figure 16 shows the summary and ANOVA tables for the model with the `year` parameter removed, notably the ANOVA table confirms there is not a significant loss of explanatory power by removing this variable. The summary table also displays three of the four coefficients now have a p-value of less than 1%.

Similar to section 5.1 we can now begin combining factors. Figure 16 tells us that there is no significant difference between factor levels for the medicine and arts faculties, so we will combine them. We have a new model, with faculty factors `MA`, `S` and `E`, fig. 17 shows us that by a likelihood ratio test, there is no significant loss of explanatory power by combining these factors together, the summary table also shows this new `MA` factor to be significantly different to the engineering faculty to a significance level of $< 5\%$. The summary table also tells us that there is no significant

difference between the factor levels in the faculties of engineering and sciences, we can combine these as we did in section 5.1. After combining these factor levels, fig. 18 shows us that all parameters are significant to $< 1\%$, as well as this, the ANOVA table shows us that a likelihood ratio test confirms there is no significant loss of explanatory power by combining the `S` and `E` factors. Alongside this we have the lowest AIC from all of the male specific models so far. For male data it appears that fig. 18 displays the best male specific model, that is the model depending only on faculty, with the merged `MA` and `SE` faculties.

6.2 Female Data

Similar to section 6.1, we will first consider a female specific model with `year` and `faculty` parameters with no interactions. Within fig. 19, more specifically the ANOVA table, we can see that the addition of the `year` parameter reduces the deviance by a negligible amount. Furthermore from the summary table we can see no parameters have a significance level of $\leq 5\%$, meaning none of the parameters currently have a significant effect on the result. This could be down to overfitting due to a high parameter number, or due to the small dataset. For our second female specific model, we have removed the `year` parameter. Figure 20 shows us that with a likelihood ratio test, there has been no significant loss in explanatory power by removing the `year` parameter, furthermore it now shows a very significant intercept. That being said, the parameter comparing females in the science and arts faculties is not significant to a 5% level, indicating it is not necessarily useful at predicting whether a female will survive for 50 years after graduation. From fig. 5 we do see a difference in observed survival proportions across the two female faculties. However, this does not establish why the faculty coefficient is not significant. The female subset contains 20 grouped observations representing 220 graduates. Nevertheless, we can investigate a third model, this model is the null model. The null model is essentially the model that assumes a females chances of survival for 50 years after graduation do not depend on any factors apart from the fact they are female, i.e. faculty nor year of graduation have an effect on survival. For the null model, fig. 21 shows us the summary table, as well as ANOVA tables for the likelihood ratio tests between the model in fig. 20 as well as the original female model in fig. 19. Both likelihood ratio tests describe how there is no significant difference to a 5% level between the null models and either of the previous 2 female models discussed. It has a higher AIC than either of the two models, however the increase is relatively small, suggesting the added complexity of including `faculty` or `year` does not substantially improve the model fit.

Overall for females, whilst there may be real differences in the survival outcomes for females based on either faculty or year of graduation, we do not have sufficient statistical evidence to make this claim at a 5% significance level. For now the null model provides the most justifiable baseline for female survival for 50 years after graduation.

7 Conclusion

This report explored the effects of faculty, sex and year of graduation on determining the probability of surviving for 50 years after graduation, using binomial models. Section 3 revealed notable patterns, particularly the effect of sex on survival probability, with females generally much more likely to survive for 50 years after graduation. It also demonstrated how those from the science and engineering faculties had higher observed 50-year survival proportions than those from the medicine or arts faculties. When considering models with all datapoints, our models confirmed these observations with sex and faculty being the most influential predictors, whilst year of graduation showed negligible significance.

In section 5.1 we investigated reducing the parameter number, more specifically removing the year variable and combining factors. We demonstrated that combining the faculties of science and

engineering as well as medicine and arts demonstrated no significant drop in explanatory power, whilst reducing parameter number. Investigating interaction effects as well as binning our `year` variable into discrete categories did not significantly improve model fit. When modelling our entire dataset, `model4` found in fig. 10 was preferred for its simplicity and adequate fit.

When separating the data by sex, the male data supported a simplified model with two faculty groups, found in fig. 18, which strikes a balance between the number of predictors and explanatory power. When considering female only data, we did not find sufficient evidence to prefer models including `faculty` or `year` over the null model at a 5% significance level.

A Appendix

```r
model1 = glm(
  cbind(survive, total - survive) ~ year + faculty + sex,
  family = binomial
)
summary(model1)
anova(model1)
```

```
Call:
glm(formula = cbind(survive, total - survive) ~ year + faculty +
    sex, family = binomial)

Coefficients:
             Estimate Std. Error z value Pr(>|z|)
(Intercept) -52.66139   48.71620  -1.081 0.279705
year          0.02784    0.02509   1.110 0.267067
facultyE      0.65193    0.24935   2.615 0.008934 **
facultyM      0.18391    0.20603   0.893 0.372042
facultyS      0.61444    0.21120   2.909 0.003623 **
sexM         -0.85542    0.22422  -3.815 0.000136 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 75.419  on 57  degrees of freedom
Residual deviance: 48.226  on 52  degrees of freedom
AIC: 205.64

Number of Fisher Scoring iterations: 4
```

```
Analysis of Deviance Table

Model: binomial, link: logit

Response: cbind(survive, total - survive)

Terms added sequentially (first to last)


        Df Deviance Resid. Df Resid. Dev
NULL                    57     75.419
year     1   2.3427         56     73.076
faculty  3   9.3641         53     63.712
sex      1  15.4856         52     48.226
```

Figure 6: Model 1 generation, summary and ANOVA table.

```r
ex_data1 = data.frame(sex = c('M', 'F'), year = c(1941, 1938), faculty =
↪  c('M', 'E'))
ex_data_fit = predict.glm(model1, ex_data1, type = 'response', se.fit = T)$fit
ex_data_se = predict.glm(model1, ex_data1, type = 'response', se.fit =
↪  T)$se.fit
cat("Fitted Values:\n", ex_data_fit, "\n",
    "Standard Errors:\n", ex_data_se, "\n")
```

```
Fitted Values:
 0.6694586 0.8749723
Standard Errors:
 0.03043977 0.03552879
```

Figure 7: Example data predictions using model 1.

```r
model2 = glm(
  cbind(survive, total - survive) ~  faculty + sex,
  family = binomial
)
summary(model2)
anova(model2, model1, test = 'Chisq')
```

```
Call:
glm(formula = cbind(survive, total - survive) ~ faculty + sex,
    family = binomial)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)   1.4073     0.1804   7.800  6.2e-15 ***
facultyE      0.6957     0.2462   2.826  0.00472 **
facultyM      0.2180     0.2036   1.071  0.28429
facultyS      0.6528     0.2083   3.133  0.00173 **
sexM         -0.8662     0.2240  -3.868  0.00011 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 75.419  on 57  degrees of freedom
Residual deviance: 49.459  on 53  degrees of freedom
AIC: 204.87

Number of Fisher Scoring iterations: 4
```

```
Analysis of Deviance Table

Model 1: cbind(survive, total - survive) ~ faculty + sex
Model 2: cbind(survive, total - survive) ~ year + faculty + sex
  Resid. Df Resid. Dev Df Deviance Pr(>Chi)
1        53     49.459
2        52     48.226  1   1.2332   0.2668
```

Figure 8: Model 2 generation, summary and ANOVA table.

```r
faculty2 = faculty
faculty2[faculty2 == 'M'] = 'MA'
faculty2[faculty2 == 'A'] = 'MA'
faculty2 = as.factor(faculty2)

model3 = glm(cbind(survive, total - survive) ~ faculty2 + sex,
             family = binomial
)
summary(model3)
anova(model3, model2, test = 'Chisq')
```

```
Call:
glm(formula = cbind(survive, total - survive) ~ faculty2 + sex,
    family = binomial)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  1.99089    0.27382   7.271 3.57e-13 ***
faculty2MA  -0.56386    0.21389  -2.636 0.008384 **
faculty2S   -0.02929    0.24352  -0.120 0.904270
sexM        -0.75413    0.19783  -3.812 0.000138 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 75.419  on 57  degrees of freedom
Residual deviance: 50.604  on 54  degrees of freedom
AIC: 204.01

Number of Fisher Scoring iterations: 4
```

```
Analysis of Deviance Table

Model 1: cbind(survive, total - survive) ~ faculty2 + sex
Model 2: cbind(survive, total - survive) ~ faculty + sex
  Resid. Df Resid. Dev Df Deviance Pr(>Chi)
1        54     50.604
2        53     49.459  1   1.1448   0.2846
```

Figure 9: Model 3 generation, summary and ANOVA table.

```r
faculty3 = faculty
faculty3[faculty3 == 'M'] = 'MA'
faculty3[faculty3 == 'A'] = 'MA'
faculty3[faculty3 == 'S'] = 'SE'
faculty3[faculty3 == 'E'] = 'SE'
faculty3 = as.factor(faculty3)

model4 = glm(cbind(survive, total - survive) ~ faculty3 + sex,
                 family = binomial
)
summary(model4)
anova(model4, model3, test = 'Chisq')
```

```
Call:
glm(formula = cbind(survive, total - survive) ~ faculty3 + sex,
    family = binomial)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)   1.4251     0.1790   7.962 1.69e-15 ***
faculty3SE    0.5457     0.1510   3.614 0.000302 ***
sexM         -0.7517     0.1968  -3.819 0.000134 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 75.419  on 57  degrees of freedom
Residual deviance: 50.619  on 55  degrees of freedom
AIC: 202.03

Number of Fisher Scoring iterations: 4
```

```
Analysis of Deviance Table

Model 1: cbind(survive, total - survive) ~ faculty3 + sex
Model 2: cbind(survive, total - survive) ~ faculty2 + sex
  Resid. Df Resid. Dev Df Deviance Pr(>Chi)
1        55     50.619
2        54     50.604  1 0.014481   0.9042
```

Figure 10: Model 4 generation, summary and ANOVA table.

```r
anova(model4, model1, test = 'Chisq')
```

```
Analysis of Deviance Table

Model 1: cbind(survive, total - survive) ~ faculty3 + sex
Model 2: cbind(survive, total - survive) ~ year + faculty + sex
  Resid. Df Resid. Dev Df Deviance Pr(>Chi)
1        55     50.619
2        52     48.226  3   2.3925    0.495
```

Figure 11: Model 4 vs Model 1 LR Test

```r
model5 = glm(cbind(survive, total-survive) ~ faculty3*sex,
             family = binomial
)
summary(model5)
anova(model4, model5, test = 'Chisq')
```

```
Call:
glm(formula = cbind(survive, total - survive) ~ faculty3 * sex,
    family = binomial)

Coefficients:
                Estimate Std. Error z value Pr(>|z|)
(Intercept)       1.3789     0.1919   7.186 6.66e-13 ***
faculty3SE        0.8403     0.5085   1.653  0.09842 .
sexM             -0.6927     0.2172  -3.189  0.00143 **
faculty3SE:sexM  -0.3260     0.5328  -0.612  0.54063
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 75.419  on 57  degrees of freedom
Residual deviance: 50.225  on 54  degrees of freedom
AIC: 203.64

Number of Fisher Scoring iterations: 4
```

```
Analysis of Deviance Table

Model 1: cbind(survive, total - survive) ~ faculty3 + sex
Model 2: cbind(survive, total - survive) ~ faculty3 * sex
  Resid. Df Resid. Dev Df Deviance Pr(>Chi)
1        55     50.619
2        54     50.225  1  0.39377   0.5303
```

Figure 12: Model 5 generation, summary and ANOVA table.

```r
pre_post_war = cut(year, breaks = c(1930, 1939, 1945, 2000), labels =
 ↪  c("Pre-War", "During-War", "Post-War"))
pre_post_war = as.factor(pre_post_war)

war_model = glm(cbind(survive, total-survive) ~ pre_post_war + faculty3 + sex,
                family = binomial)
summary(war_model)
anova(model4, war_model, test = 'Chisq')
```

```
Call:
glm(formula = cbind(survive, total - survive) ~ pre_post_war +
    faculty3 + sex, family = binomial)

Coefficients:
                        Estimate Std. Error z value Pr(>|z|)
(Intercept)              1.41748    0.23485   6.036 1.58e-09 ***
pre_post_warDuring-War  -0.03784    0.19211  -0.197 0.843838
pre_post_warPost-War     0.35648    0.23876   1.493 0.135432
faculty3SE               0.55951    0.15273   3.663 0.000249 ***
sexM                    -0.80375    0.19888  -4.041 5.32e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 75.419  on 57  degrees of freedom
Residual deviance: 46.182  on 53  degrees of freedom
AIC: 201.59

Number of Fisher Scoring iterations: 4
```

```
Analysis of Deviance Table

Model 1: cbind(survive, total - survive) ~ faculty3 + sex
Model 2: cbind(survive, total - survive) ~ pre_post_war + faculty3 + sex
  Resid. Df Resid. Dev Df Deviance Pr(>Chi)
1        55     50.619
2        53     46.182  2   4.4364   0.1088
```

Figure 13: War binning model generation, summary and ANOVA table.

```r
decade_split = cut(year, breaks = c(1930, 1939, 2000), labels = c("Pre-1940",
↪  "Post-1940"))
decade_split = as.factor(decade_split)

decade_model = glm(cbind(survive, total-survive) ~ decade_split + faculty3 +
↪  sex,
                   family = binomial)
summary(decade_model)
anova(model4, decade_model, test = 'Chisq')
```

```
Call:
glm(formula = cbind(survive, total - survive) ~ decade_split +
    faculty3 + sex, family = binomial)

Coefficients:
                       Estimate Std. Error z value Pr(>|z|)
(Intercept)             1.37754    0.23332   5.904 3.54e-09 ***
decade_splitPost-1940   0.05912    0.18660   0.317 0.751395
faculty3SE              0.54021    0.15201   3.554 0.000380 ***
sexM                   -0.75030    0.19691  -3.810 0.000139 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 75.419  on 57  degrees of freedom
Residual deviance: 50.519  on 54  degrees of freedom
AIC: 203.93

Number of Fisher Scoring iterations: 4
```

```
Analysis of Deviance Table

Model 1: cbind(survive, total - survive) ~ faculty3 + sex
Model 2: cbind(survive, total - survive) ~ decade_split + faculty3 + sex
  Resid. Df Resid. Dev Df Deviance Pr(>Chi)
1        55     50.619
2        54     50.519  1 0.099841    0.752
```

Figure 14: Decade binning model generation, summary and ANOVA table.

```r
male_data = data[data$sex == "M", ]
colnames(male_data) <- paste0(colnames(male_data), "_M")
attach(male_data)
as.factor(faculty_M)

male_model1 = glm(cbind(survive_M, total_M-survive_M) ~ faculty_M + year_M,
                  family = binomial
)
summary(male_model1)
anova(male_model1)
```

```
Call:
glm(formula = cbind(survive_M, total_M - survive_M) ~ faculty_M +
    year_M, family = binomial)

Coefficients:
             Estimate Std. Error z value Pr(>|z|)
(Intercept) -85.07485   52.87174  -1.609   0.1076
faculty_ME    0.60525    0.25561   2.368   0.0179 *
faculty_MM    0.14196    0.21344   0.665   0.5060
faculty_MS    0.54955    0.23420   2.347   0.0189 *
year_M        0.04411    0.02723   1.620   0.1053
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 48.343  on 37  degrees of freedom
Residual deviance: 34.137  on 33  degrees of freedom
AIC: 157.79

Number of Fisher Scoring iterations: 4
```

```
Analysis of Deviance Table

Model: binomial, link: logit

Response: cbind(survive_M, total_M - survive_M)

Terms added sequentially (first to last)


          Df Deviance Resid. Df Resid. Dev
NULL                         37     48.343
faculty_M  3  11.5749         34     36.768
year_M     1   2.6312         33     34.137
```

Figure 15: Male model 1 generation, summary and ANOVA table.

```r
male_model2 = glm(cbind(survive_M, total_M-survive_M) ~ faculty_M,
                  family = binomial
)
summary(male_model2)
anova(male_model2, male_model1, test = "Chisq")
```

```
Call:
glm(formula = cbind(survive_M, total_M - survive_M) ~ faculty_M,
    family = binomial)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)   0.5621     0.1660   3.387 0.000707 ***
faculty_ME    0.6746     0.2518   2.680 0.007370 **
faculty_MM    0.1970     0.2103   0.937 0.348920
faculty_MS    0.6120     0.2307   2.653 0.007977 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 48.343  on 37  degrees of freedom
Residual deviance: 36.768  on 34  degrees of freedom
AIC: 158.42

Number of Fisher Scoring iterations: 4
```

```
Analysis of Deviance Table

Model 1: cbind(survive_M, total_M - survive_M) ~ faculty_M
Model 2: cbind(survive_M, total_M - survive_M) ~ faculty_M + year_M
  Resid. Df Resid. Dev Df Deviance Pr(>Chi)
1        34     36.768
2        33     34.137  1   2.6312   0.1048
```

Figure 16: Male model 2 generation, summary and ANOVA table.

```r
faculty2_M = faculty_M
faculty2_M[faculty2_M %in% c('M', 'A')] = 'MA'
faculty2_M = as.factor(faculty2_M)

male_model3 = glm(cbind(survive_M, total_M-survive_M) ~ faculty2_M,
                  family = binomial
)
summary(male_model3)
anova(male_model3, male_model2, test = "Chisq")
```

```
Call:
glm(formula = cbind(survive_M, total_M - survive_M) ~ faculty2_M,
    family = binomial)

Coefficients:
             Estimate Std. Error z value Pr(>|z|)
(Intercept)   1.23676    0.18932   6.533 6.46e-11 ***
faculty2_MMA -0.55054    0.21497  -2.561   0.0104 *
faculty2_MS  -0.06264    0.24801  -0.253   0.8006
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 48.343  on 37  degrees of freedom
Residual deviance: 37.642  on 35  degrees of freedom
AIC: 157.3

Number of Fisher Scoring iterations: 4
```

```
Analysis of Deviance Table

Model 1: cbind(survive_M, total_M - survive_M) ~ faculty2_M
Model 2: cbind(survive_M, total_M - survive_M) ~ faculty_M
  Resid. Df Resid. Dev Df Deviance Pr(>Chi)
1        35     37.642
2        34     36.768  1  0.87373   0.3499
```

Figure 17: Male model 3 generation, summary and ANOVA table.

```r
faculty3_M = faculty_M
faculty3_M[faculty3_M %in% c('M', 'A')] = "MA"
faculty3_M[faculty3_M %in% c('S', 'E')] = "SE"
faculty3_M = as.factor(faculty3_M)
male_model4 = glm(cbind(survive_M, total_M-survive_M) ~ faculty3_M,
                                family = binomial
)
summary(male_model4)
anova(male_model4, male_model3, test = "Chisq")
```

```
Call:
glm(formula = cbind(survive_M, total_M - survive_M) ~ faculty3_M,
    family = binomial)

Coefficients:
             Estimate Std. Error z value Pr(>|z|)
(Intercept)    0.6862     0.1018   6.739 1.59e-11 ***
faculty3_MSE   0.5143     0.1591   3.232  0.00123 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 48.343  on 37  degrees of freedom
Residual deviance: 37.706  on 36  degrees of freedom
AIC: 155.36

Number of Fisher Scoring iterations: 4
```

```
Analysis of Deviance Table

Model 1: cbind(survive_M, total_M - survive_M) ~ faculty3_M
Model 2: cbind(survive_M, total_M - survive_M) ~ faculty2_M
  Resid. Df Resid. Dev Df Deviance Pr(>Chi)
1        36     37.706
2        35     37.642  1 0.063917   0.8004
```

Figure 18: Male model 4 generation, summary and ANOVA table.

```r
female_data = data[data$sex == "F", ]
colnames(female_data) <- paste0(colnames(female_data), "_F")
attach(female_data)
as.factor(faculty_F)

female_model1 = glm(cbind(survive_F, total_F-survive_F) ~ faculty_F + year_F,
                    family = binomial
)
summary(female_model1)
anova(female_model1)
```

```
Call:
glm(formula = cbind(survive_F, total_F - survive_F) ~ faculty_F +
    year_F, family = binomial)

Coefficients:
             Estimate Std. Error z value Pr(>|z|)
(Intercept) 125.50767  126.34925   0.993   0.3205
faculty_FS    0.91401    0.51434   1.777   0.0756 .
year_F       -0.06391    0.06505  -0.983   0.3259
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 15.695  on 19  degrees of freedom
Residual deviance: 11.549  on 17  degrees of freedom
AIC: 49.306

Number of Fisher Scoring iterations: 4
```

```
Analysis of Deviance Table

Model: binomial, link: logit

Response: cbind(survive_F, total_F - survive_F)

Terms added sequentially (first to last)


          Df Deviance Resid. Df Resid. Dev
NULL                         19     15.695
faculty_F  1   3.1765         18     12.519
year_F     1   0.9702         17     11.549
```

Figure 19: Female model 1 generation, summary and ANOVA table.

```r
female_model2 = glm(cbind(survive_F, total_F-survive_F) ~ faculty_F,
                    family = binomial
)
summary(female_model2)
anova(female_model2, female_model1, test = "Chisq")
```

```
Call:
glm(formula = cbind(survive_F, total_F - survive_F) ~ faculty_F,
    family = binomial)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)   1.3789     0.1919   7.186 6.66e-13 ***
faculty_FS    0.8403     0.5085   1.653   0.0984 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 15.695  on 19  degrees of freedom
Residual deviance: 12.519  on 18  degrees of freedom
AIC: 48.276

Number of Fisher Scoring iterations: 4
```

```
Analysis of Deviance Table

Model 1: cbind(survive_F, total_F - survive_F) ~ faculty_F
Model 2: cbind(survive_F, total_F - survive_F) ~ faculty_F + year_F
  Resid. Df Resid. Dev Df Deviance Pr(>Chi)
1        18     12.519
2        17     11.549  1  0.97019   0.3246
```

Figure 20: Female model 2 generation, summary and ANOVA table.

```r
female_model3 = glm(cbind(survive_F, total_F-survive_F) ~ 1,
                    family = binomial
)
summary(female_model3)
anova(female_model3, female_model2, test = "Chisq")
anova(female_model3, female_model1, test = "Chisq")

```

```
Call:
glm(formula = cbind(survive_F, total_F - survive_F) ~ 1, family = binomial
    )

Coefficients:
             Estimate Std. Error z value Pr(>|z|)
(Intercept)   1.5349     0.1765   8.695   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 15.695  on 19  degrees of freedom
Residual deviance: 15.695  on 19  degrees of freedom
AIC: 49.453

Number of Fisher Scoring iterations: 4
```

```
Analysis of Deviance Table

Model 1: cbind(survive_F, total_F - survive_F) ~ 1
Model 2: cbind(survive_F, total_F - survive_F) ~ faculty_F
  Resid. Df Resid. Dev Df Deviance Pr(>Chi)
1         19     15.695
2         18     12.519  1   3.1765   0.0747 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Analysis of Deviance Table

Model 1: cbind(survive_F, total_F - survive_F) ~ 1
Model 2: cbind(survive_F, total_F - survive_F) ~ faculty_F + year_F
  Resid. Df Resid. Dev Df Deviance Pr(>Chi)
1         19     15.695
2         17     11.549  2   4.1467   0.1258
```

Figure 21: Female model 3 generation, summary and ANOVA table.